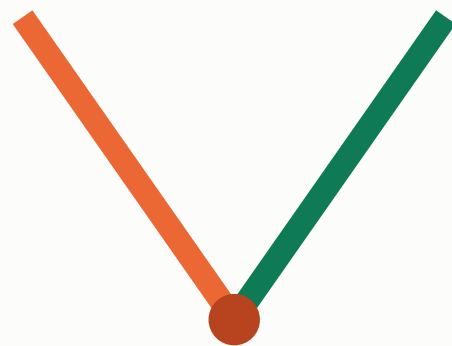




SWITCHBACK 人字轨

Centennial Jing-Zhang AI Innovation Belt · A3 booklet

Cindy-Master · package_type=professional_design_package

Overall design area 11.4 km2 - Ascent, Reversing Point, Departure, closed by the two return wings

01 Concept and naming system

Centennial Jing-Zhang AI Innovation Belt · A3 booklet

Cindy-Master · package_type=professional_design_package

The most famous engineering legacy of the Jing-Zhang railway is the switchback at Qinglongqiao: where the gradient could not be climbed directly, the train reversed once and traded a change of direction for height. That century-old logic is structurally identical to how AI capability actually accumulates - not by a straight sprint, but by reversing repeatedly through research, application, feedback and research again.

The proposal translates this into spatial structure. Zhongzhiyuan in the north is the ascent, doing the 0-to-1 full-stack work. The AI Origin Community in the middle is the reversing point, turning results into open-source release and talent gathering. Dazhongsi in the south is the departure, doing the 1-to-N market and international work. The Zhongguancun and Xiaoyuehe wings are the return arms that send capital and real scenario data back to the ascent, so the belt is a closed loop rather than a one-way corridor.

The logo is built from two reversing strokes and reads three ways at once: the character 人 (human-centred governance), a railway turnout symbol (the Jing-Zhang legacy) and a feedback-loop arrow (the iterative logic of AI). The naming system uses railway operating grammar so that every object carries its position in the loop inside its own name.

Naming system

Belt	SWITCHBACK 人字轨
Ascent	Zhongzhiyuan AI Acceleration Area
Reversing point	Beijing AI Origin Community
Departure	Dazhongsi AI Industry Cluster
West arm	Zhongguancun technology-service wing - factor return
East arm	Xiaoyuehe scenario-empowerment wing - scenario return

Positioning and functions

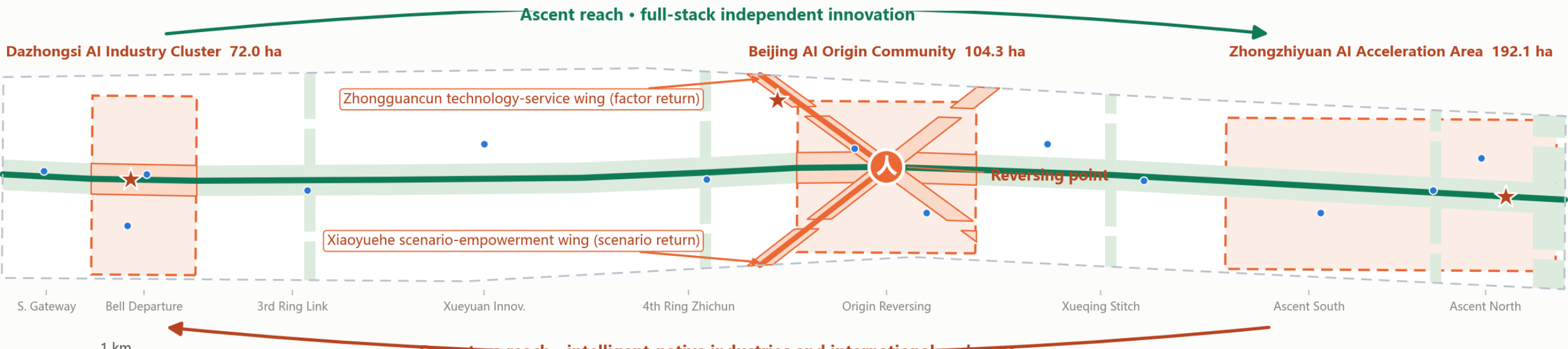
Three positionings: centennial Jing-Zhang cultural belt / urban AI living-experience belt / AI convergence-innovation belt

Five functions: full-stack independent AI innovation / world-class AI ecosystem / AI+ scenario empowerment / intelligent vibrant AI city / a global voice in AI governance

SWITCHBACK — an AI innovation loop that gains height by reversing

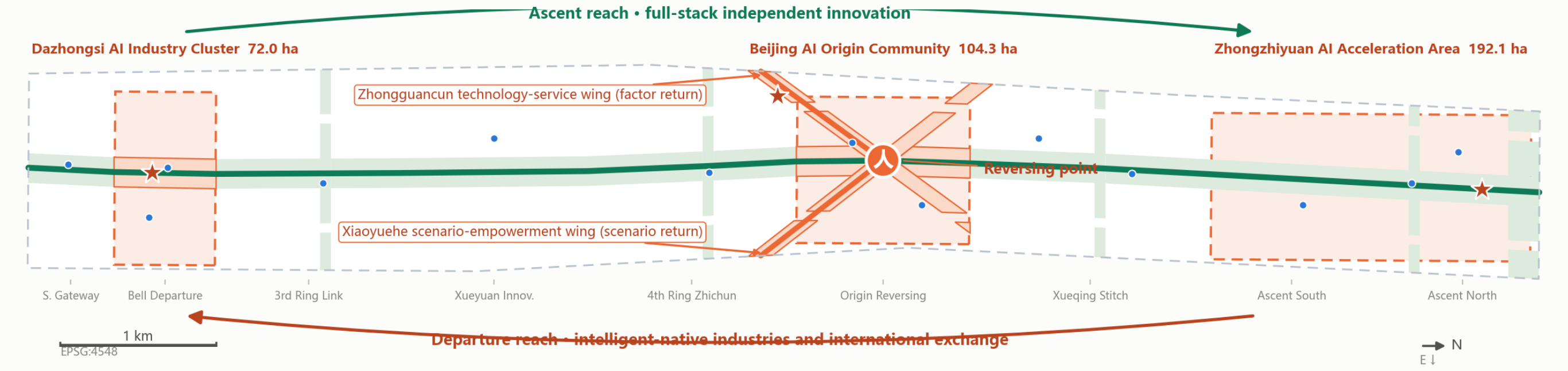
SWITCHBACK 人字轨 · Cindy-Master

Centennial Jing-Zhang AI Belt · overall design area 11.4 km² · Ascent — Reversing Point — Departure, closed by two return wings



SWITCHBACK — an AI innovation loop that gains height by reversing

Centennial Jing-Zhang AI Belt · overall design area 11.4 km² · Ascent — Reversing Point — Departure, closed by two return wings



Naming system and logo direction

人字轨 SWITCHBACK

Two reversing strokes form 人 (human): the Jing-Zhang switchback, a railway turnout symbol and a feedback-loop arrow.

Belt	SWITCHBACK 人字轨
Ascent	Zhongzhiyuan
Reversing	AI Origin Community
Departure	Dazhongsi
West arm	Zhongguancun service wing
East arm	Xiaoyuehe scene wing

Three positionings and five functions

Centennial Jing-Zhang cultural belt

Urban AI living-experience belt

AI convergence-innovation belt

- Full-stack independent AI innovation
- World-class AI innovation ecosystem
- AI+ scenario empowerment
- Intelligent, vibrant AI city
- A global voice in AI governance

Return loop: three areas, two wings

- » The ascent reach does the 0-to-1 full-stack work and accumulates capability northward;
- » the reversing point changes direction: open-source release, transfer to market, talent;
- » the departure reach does the 1-to-N market and international work;
- » the two wings return capital and real scenario data to the ascent, closing the loop.

Legend and precision statement

- Green spine (slow-mobility trunk)
- Switchback arms — public space
- Park and protective green space
- Squares and public space
- Key areas (provisional_rough)
- ★ AI pilgrimage landmarks
- AI scenario nodes
- Overall design area (provisional substitute boundary)

Precision statement: the exact official SITE_BOUNDARY and KEY_AREA polygons are not public. The extents shown are provisional substitutes and must not serve as an official redline, an approval basis or a precise-area basis; every layer and metric is recomputed once official data is released.

Sources: the brief/site-package/ site package and the announcement's textual extents; geometry and metrics are recomputed from this package's geometry/*.geojson.

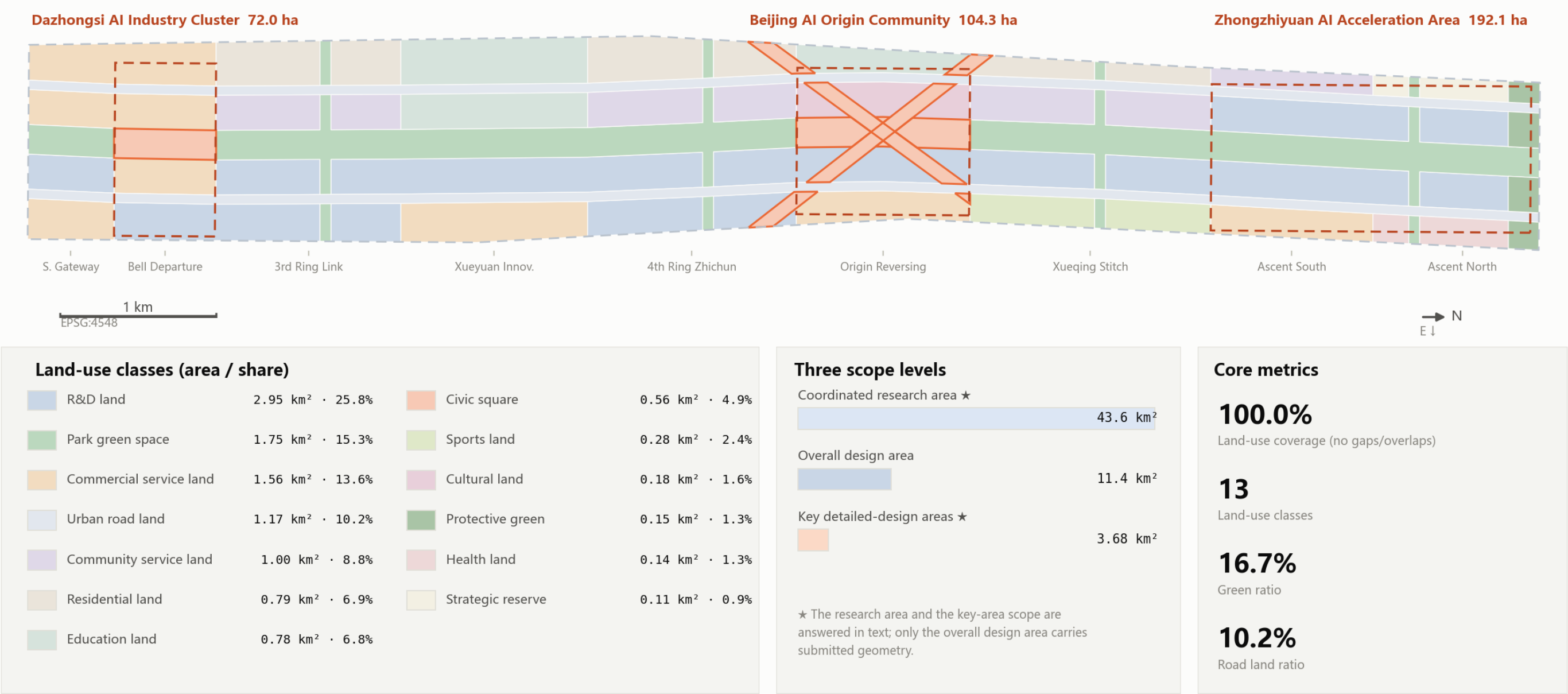
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Status: an open co-creation concept and reference scheme for professional teams to deepen. It does not replace statutory planning and is not a government-approved conclusion.

Land-use structure and the three-level scope framework

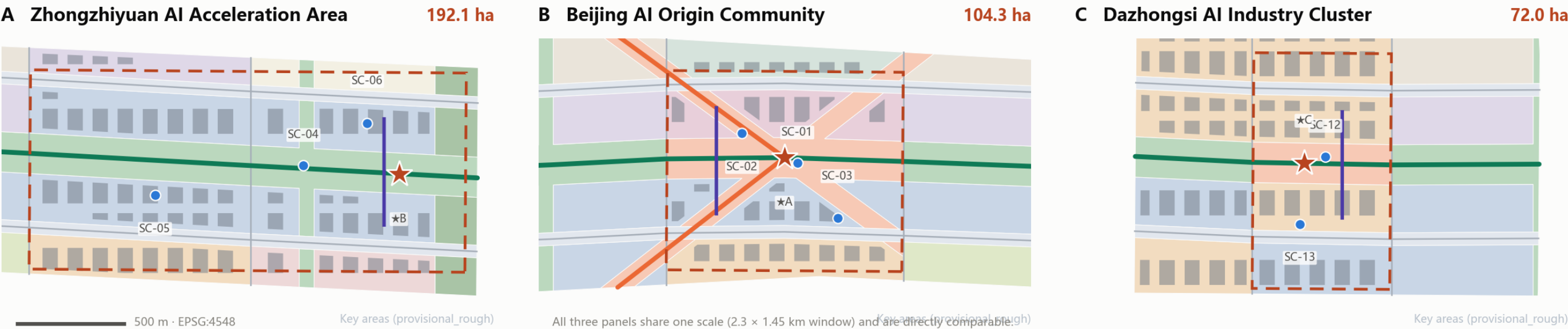
SWITCHBACK 人字轨 · Cindy-Master

A complete land-use partition around the green spine (no gaps, no overlaps); areas recomputed in EPSG:4548



Index of the three key detailed-design areas

Same-scale small multiples: Zhongzhiyuan AI Acceleration Area · Beijing AI Origin Community · Dazhongsi AI Industry Cluster



• Role: ascent reach — full-stack independent innovation and a voice in AI governance • Moves: an open test loop on the Qinghe frontage, a standards-governance display axis, a low-carbon exchange loop • Scenarios: SC-04 model test ground, SC-05 safety red-team sandbox, SC-06 standards workshop • Landmark: ★B The Ascent Beacon (compute and governance display) • Dependencies: Qinghe blue-line and flood conditions, test-ground safety and regulatory authorisation, external traffic review	• Role: reversing point — transfer to market, open-source release and talent gathering • Moves: the Reversing Plaza plus the two arms stitching campus, park and neighbourhood • Scenarios: SC-01 open-source release hall, SC-02 campus-to-market street, SC-03 talent one-stop point • Landmark: ★A The Reversing Marker (open-source contributor honour system) • Dependencies: campus and park ownership edges, ground-floor and property coordination, station integration	• Role: departure reach — intelligent-native industries and international exchange • Moves: four-quadrant pedestrian connection at the station, the Bell Platform plaza, frontage renewal • Scenarios: SC-12 consumption block, SC-13 data-element reception hall, SC-14 roadshow hall • Landmark: ★C The Bell Platform (annual release and bell-ringing ceremony) • Dependencies: station exits and utilities, junction channelisation and traffic safety, heritage and tenure review
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Sources: the brief/site-package/ site package and the announcement’s textual extents; geometry and metrics are recomputed from this package’s geometry/*_geojson.

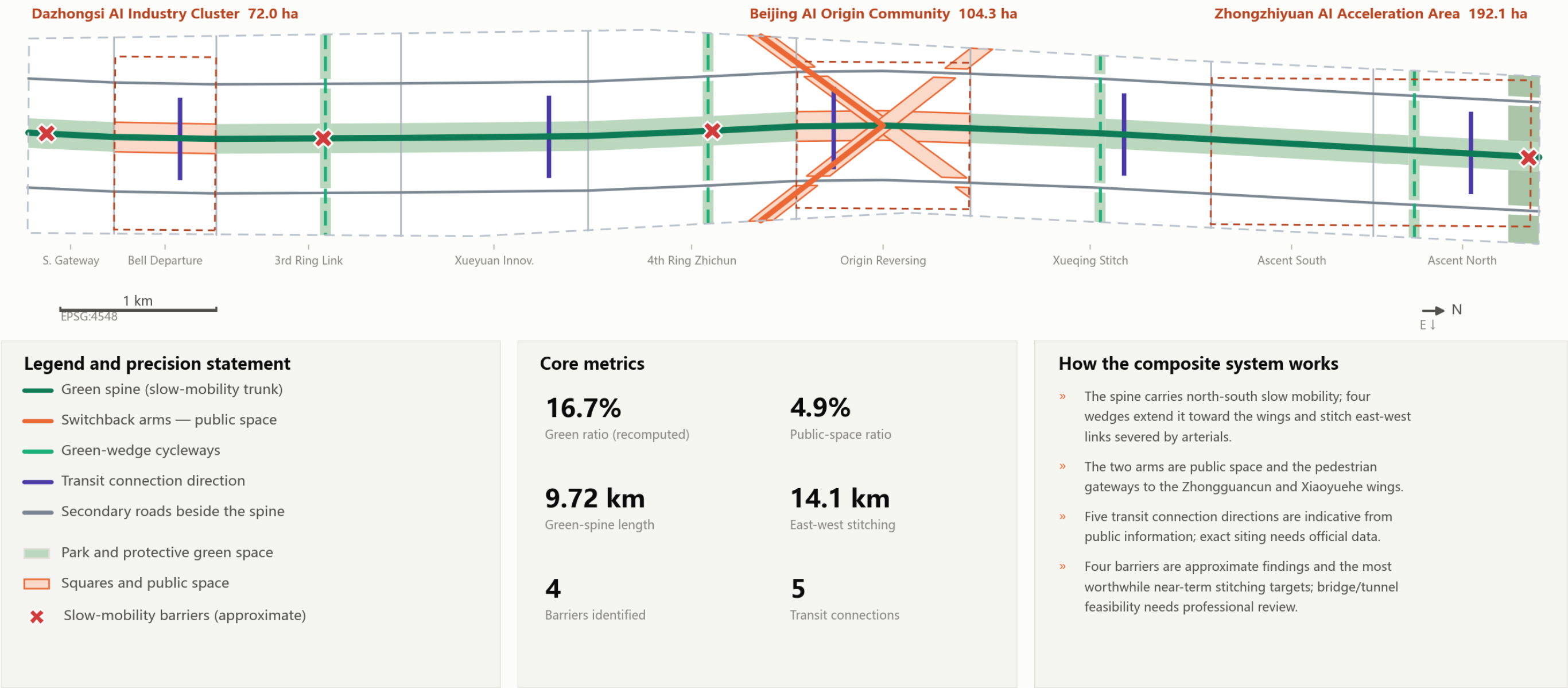
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Mobility and blue-green public space as one composite system

SWITCHBACK 人字轨 · Cindy-Master

A continuous north-south green spine + four transverse wedges + the two switchback arms



Sources: the brief/site-package/ site package and the announcement’s textual extents; geometry and metrics are recomputed from this package’s geometry/* .geojson.
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06 AI scenario cards and personas

Centennial Jing-Zhang AI Innovation Belt · A3 booklet

Scenario card	Type	Spatial host	Privacy boundary	Human review
SC-01 Open-Source Release Hall	AI+industry services	AI Origin Community · Reversing Plaza	No movement tracking; honour display is revocable by the person	Content reviewed by a community maintainer and a venue operator
SC-02 Campus-to-Market Transfer Street	AI+education and transfer	AI Origin Community · campus frontage	No access to internal campus systems; personal display requires consent	Transfer advice is advisory and must be reviewed by professional institutions
SC-03 Talent One-Stop Service Point	AI+living services	AI Origin Community · east service frontage	Documents stored locally and deleted on completion; no cross-agency profiling	Every conclusion is confirmed by a human officer
SC-04 Full-Stack Model Test Ground	Industry test	Zhongzhiyuan · ascent test loop	Physically separated test area; external data does not leave the site	Results are published only after third-party verification
SC-05 Safety-Governance Red-Team Sandbox	Industry test	Zhongzhiyuan · standards display axis	Fully logged, auditable and revocable	High-risk tests need an observer present with a human stop authority
SC-06 Standards and Governance Workshop	AI-governance voice	Zhongzhiyuan · ascent north	No personal data captured in the public observation area	Conclusions follow the standards bodies' formal procedures
SC-07 Low-Speed Robot Delivery Loop	Industry test	Xueqing reach · spine test loop	No face recognition; imagery processed locally with limited retention	Remote human takeover and physical e-stop; incidents must be reported
SC-08 AI+Community Health Navigation	AI+health	Xueqing reach · community frontage	No symptom or identity storage; navigation and booking hand-off only	No diagnosis; always routed to a licensed clinician
SC-09 Accessible Slow-Mobility Guidance	AI+mobility	4th Ring Zhichun reach · green wedge	Routing computed on-device; no personal trips uploaded	Accessibility claims are field-verified by disabled user representatives
SC-10 Xueyuan Road AI Learning Point	AI+education	Xueyuan Road reach · education frontage	No personal profiles for minors; parents can review and delete	Teachers lead; AI assists only
SC-11 AI+Legal and Compliance Desk	AI+legal	3rd Ring Link reach · service frontage	Consultations are not retained and are not used for training	Information retrieval only; legal opinions come from licensed lawyers
SC-12 Intelligent-Native Consumption Block	AI+consumption	Dazhongsi · departure frontage	No individual identification by default; personalisation is opt-in and revocable	Pricing and recommendation must be explainable and appealable
SC-13 Data-Element Reception Hall	AI+data elements	Dazhongsi · east industry frontage	Raw data is never delivered on site	Every match is logged and traceable to its authorisation
SC-14 South Gateway International Roadshow Hall	International exchange	South Gateway reach	Visitor information is handled separately under protocol rules	External messaging distinguishes submitted, reviewed, selected and implemented

07 AI landmarks and cultural narrative

Centennial Jing-Zhang AI Innovation Belt · A3 booklet

Landmark	Location	Design intent
★A The Reversing Marker	Reversing Plaza, AI Origin Community	A public structure derived from the switchback geometry, whose surface is an updatable contributor honour interface recording open-source contributions and Agent proposals entering the public knowledge base. Honour data comes only from public repository records, is removable on request, and shows no personal identity data.
★B The Ascent Beacon	Ascent north reach, Zhongzhiyuan	A display structure explaining compute, energy use and standards governance to the public, turning normally invisible infrastructure into a readable urban object. Displayed data must be verifiable public or authorised data, with method and date stated.
★C The Bell Platform	Bell departure reach, Dazhongsi	A release platform echoing the Dazhongsi place name and its bell imagery, serving as the ceremonial public space for annual releases and international roadshows. Kept clear of the heritage monument; form, material and lighting require heritage and character review.
★D The Switchback Junction Deck	Junction of the west arm and the green spine	A viewing and reversing deck across a slow-mobility barrier; the one elevated point from which the ascent-reverse-departure structure can be read as a whole. Crossing form, structure and engineering feasibility require separate professional study; only position and function are proposed here.

Cultural narrative: the value of the Jing-Zhang railway is not the track but the attitude it records - finding a solution under constraint. Zhongguancun's innovation culture and the new AI culture continue the same line. The proposal therefore suggests not a railway theme park but a heritage frontage designed as a continuously updatable narrative carrier.

08 Renewal projects and phasing

Centennial Jing-Zhang AI Innovation Belt · A3 booklet

ID	Project	Type	Phase	Dependencies
JZ-01	Green-spine trunk continuity and heritage frontage tidy-up	Public space	Near term	Heritage tenure and protection requirements, green and blue-line conditions
JZ-02	Reversing Plaza and the two switchback arms	Public space	Near term	Campus and park tenure edges, underground utilities
JZ-03	Open-source release hall and the Reversing Marker honour system	Industry service / operation	Near term	Operator, content-governance rules, rights clearance
JZ-04	Four-quadrant pedestrian connection at Dazhongsi station	Station integration / slow mobility	Near term	Station exits, junction channelisation, utilities, safety assessment
JZ-05	Bell Platform plaza and frontage renewal	Urban renewal	Near term	Heritage review, retail tenure, night-noise management
JZ-06	Zhongzhiyuan open test loop and edge-compute stations	New infrastructure / industry	Mid term	Energy and compute capacity, safety and regulatory authorisation
JZ-07	Qinghe innovation frontage and low-carbon exchange loop	Blue-green space	Mid term	River blue line, ecological and flood conditions
JZ-08	South-gateway roadshow hall and gateway green	International exchange / public space	Mid term	Land conditions, protocol requirements
JZ-09	Four transverse green wedges and cycleway stitching	Slow mobility / ecology	Long term	Arterial crossing method, road redlines, engineering feasibility
JZ-10	Community-service embedding in the Xueyuan and Xueqing reaches	Public service	Long term	Existing buildings and tenure, community consent
JZ-11	Secondary roads and micro-circulation beside the spine	Mobility	Long term	Road redlines, traffic impact assessment, parking and cycle organisation
JZ-12	Switchback Junction Deck and barrier crossings	Slow mobility / engineering	Long term	Bridge/tunnel feasibility, structural and safety study (separate professional work required)



09 Global event system and long-term operation

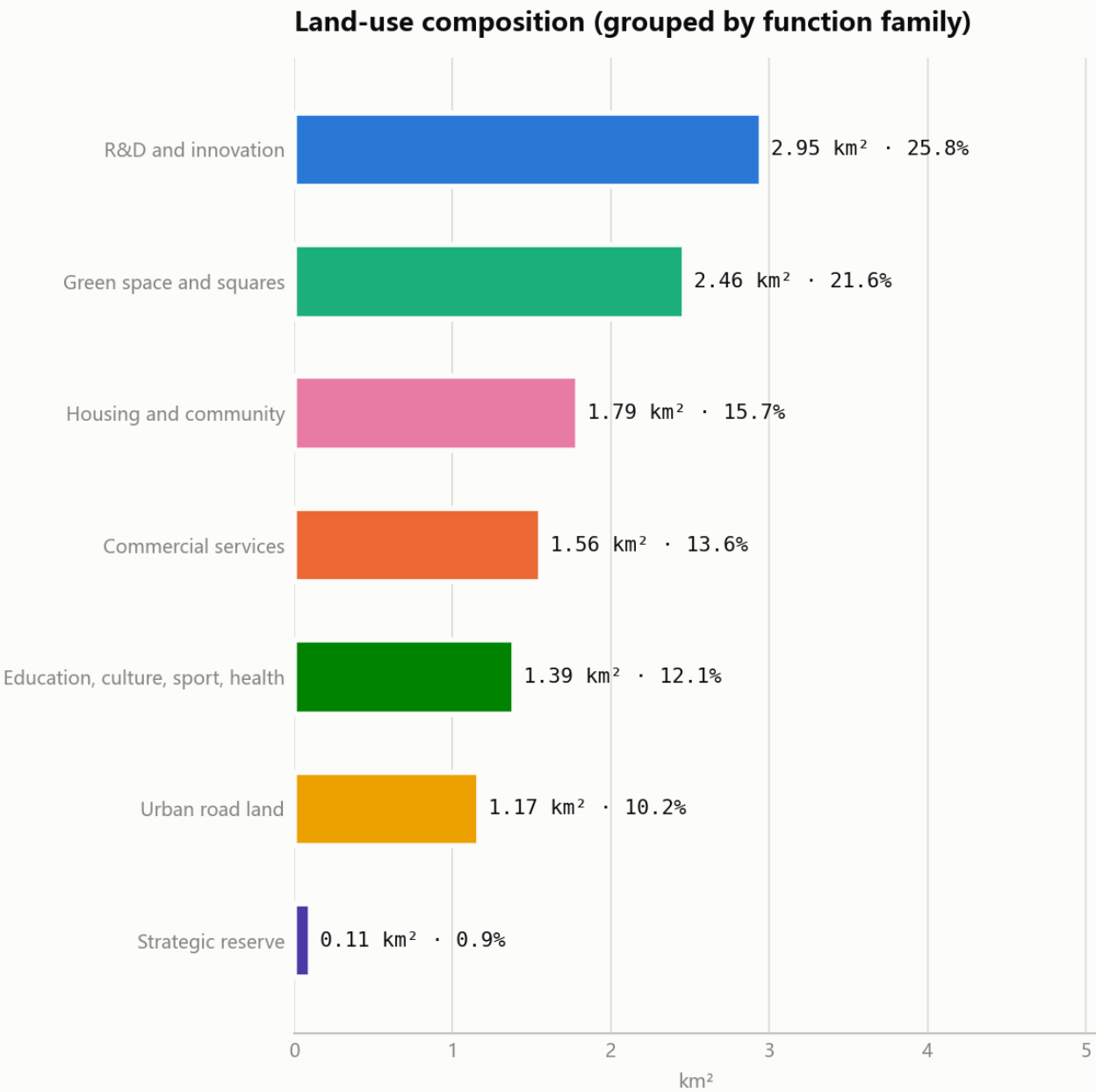
Centennial Jing-Zhang AI Innovation Belt · A3 booklet

Event	Cadence and venue	Operation and conversion
SWITCHBACK Week	Annual, centred on the Reversing Plaza	The belt's annual flagship: open-source releases, an open day on the pilgrimage route, public standards and governance debate, and a concurrent international roadshow.
Reversing Day	Quarterly	A handover day passing ascent-reach results to the departure reach's market and international channels, giving conversion a fixed rhythm.
Scenario Open Day	Monthly, rotating through the scenario cards	Real urban scenarios opened for test applications by card, with published rules, privacy boundaries and human-review requirements.
Developer Residency	Rolling year-round	Short residencies and desk support for open-source and Agent contributors, with outputs entering the public knowledge base.

Conversion pathway: residencies and open days attract developers; Reversing Day connects results to companies and capital; the talent one-stop point solves settling in; the honour system accumulates reputation; reputation attracts the next cohort. Every link needs a named responsible party and a verifiable public record. All investment, funding and policy arrangements are suggestions only.

Metric recomputation and the evidence chain

Every known metric is recomputed from the submitted GeoJSON in EPSG:4548; unknown metrics carry a reason and a precondition



Core metrics

11.41 km²

Design area (recomputed)

100.0%

Land-use coverage

16.7%

Green ratio

4.9%

Public-space ratio

14

AI scenario cards

4

AI landmarks

Metrics pending official data

- total_floor_area_sqm
- floor_area_ratio
- official_building_density
- building_height_m
- setback_m

Reasons and preconditions are recorded in metrics.json and assumptions.json; the narrative always says "pending official data".

Evidence chain

geometry/*.geojson

9 layers · topology and schema checks passed

metrics.json

56 metrics · 51 known / 5 unknown

compliance / standard / design_depth matrix

task · standard · design-depth coverage

proposal.md + proposal.en.md

primary human-readable proposal (bilingual)

report / visual / drawings

offline HTML and A3/A0 booklets (bilingual)

Structured data is the authoritative layer;

drawings and HTML are the presentation layer. Any

layer change re-runs recomputation and self-check.

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11 Risk, missing data and compliance boundary

Centennial Jing-Zhang AI Innovation Belt · A3 booklet

- Gap 1: the official design-area redline and exact key-area polygons are not public; every extent here is a provisional substitute.
- Gap 2: regulatory FAR, height, density and setback conditions are missing; related metrics are recorded as unknown.
- Gap 3: existing building quality, structural safety and tenure data are missing; only a method, not a conclusion, is given for retain-renovate-demolish.
- Gap 4: utility, energy-load, drainage, flood and fire conditions are missing; capacity conclusions are preconditions for deepening.
- Gap 5: heritage protection zones and construction control belts are unavailable; landmark and frontage design needs heritage review.
- Gap 6: exact station siting and exits are unavailable; connection segments express direction, not alignment.

Core metrics (recomputed in EPSG:4548)

11.41 km²

Design area (recomputed)

100.0%

Land-use coverage

16.7%

Green ratio

4.9%

Public-space ratio

10.2%

Road land ratio

9.72 km

Green-spine length

14

AI scenario cards

4

AI landmarks

Metrics pending official data

total_floor_area_sqm, floor_area_ratio, official_building_density, building_height_m, setback_m